



An SVD in Spherical Surface Wave Tomography

Michael Quellmalz

(joint work with Ralf Hielscher and Daniel Potts)

Chemnitz University of Technology
Faculty of Mathematics

New Trends in Parameter Identification for Mathematical Models
Chemnitz Symposium on Inverse Problems on Tour
Rio de Janeiro, 2 November 2017

Content

1. Introduction

Motivation

2. Arc transform

Definition

Singular value decomposition

3. Special cases

Arcs starting in a fixed point

Recovery of local functions

Arcs with fixed length

Content

1. Introduction

Motivation

2. Arc transform

Definition

Singular value decomposition

3. Special cases

Arcs starting in a fixed point

Recovery of local functions

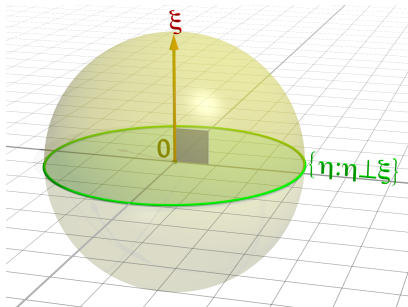
Arcs with fixed length

Funk–Radon transform

- Sphere $\mathbb{S}^2 = \{\xi \in \mathbb{R}^3 : \|\xi\| = 1\}$
- Function $f: \mathbb{S}^2 \rightarrow \mathbb{C}$
- **Funk–Radon transform** (a.k.a. Funk transform or spherical Radon transform)

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) d\lambda(\eta)$$



Theorem

[Funk 1911]

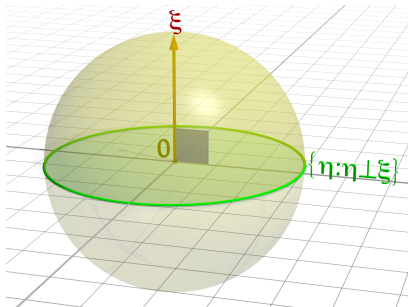
Any even function f can be reconstructed from $\mathcal{F}f$.

Funk–Radon transform

- ▶ Sphere $\mathbb{S}^2 = \{\xi \in \mathbb{R}^3 : \|\xi\| = 1\}$
- ▶ Function $f: \mathbb{S}^2 \rightarrow \mathbb{C}$
- ▶ **Funk–Radon transform** (a.k.a. Funk transform or spherical Radon transform)

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) d\lambda(\eta)$$



Theorem

[Funk 1911]

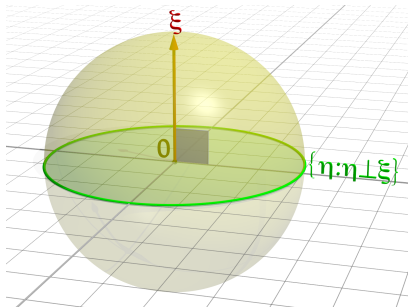
Any even function f can be reconstructed from $\mathcal{F}f$.

Funk–Radon transform

- ▶ Sphere $\mathbb{S}^2 = \{\xi \in \mathbb{R}^3 : \|\xi\| = 1\}$
- ▶ Function $f: \mathbb{S}^2 \rightarrow \mathbb{C}$
- ▶ **Funk–Radon transform** (a.k.a. Funk transform or spherical Radon transform)

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) d\lambda(\eta)$$



Theorem

[Funk 1911]

Any even function f can be reconstructed from $\mathcal{F}f$.

Spherical surface wave tomography

- ▶ Seismic waves propagate along the surface of the earth
- ▶ Speed of propagation depends on the position on S^2

Method

- ▶ Measure the traveltimes of surface waves between many pairs of epicenter and detector
- ▶ Reconstruct the local speed of propagation

Assumption

A wave propagates along the arc of a great circle.

Spherical surface wave tomography

- ▶ Seismic waves propagate along the surface of the earth
- ▶ Speed of propagation depends on the position on $\mathbb{S}^2$

Method

- ▶ Measure the traveltimes of surface waves between many pairs of epicenter and detector
- ▶ Reconstruct the local speed of propagation

Assumption

A wave propagates along the arc of a great circle.

Spherical surface wave tomography

- ▶ Seismic waves propagate along the surface of the earth
- ▶ Speed of propagation depends on the position on $\mathbb{S}^2$

Method

- ▶ Measure the traveltimes of surface waves between many pairs of epicenter and detector
- ▶ Reconstruct the local speed of propagation

Assumption

A wave propagates along the arc of a great circle.

Spherical surface wave tomography

- ▶ Seismic waves propagate along the surface of the earth
- ▶ Speed of propagation depends on the position on $\mathbb{S}^2$

Method

- ▶ Measure the traveltimes of surface waves between many pairs of epicenter and detector
- ▶ Reconstruct the local speed of propagation

Assumption

A wave propagates along the arc of a great circle.

Spherical surface wave tomography

- ▶ Seismic waves propagate along the surface of the earth
- ▶ Speed of propagation depends on the position on $\mathbb{S}^2$

Method

- ▶ Measure the traveltimes of surface waves between many pairs of epicenter and detector
- ▶ Reconstruct the local speed of propagation

Assumption

A wave propagates along the arc of a great circle.

Spherical surface wave tomography

- ▶ Seismic waves propagate along the surface of the earth
- ▶ Speed of propagation depends on the position on $\mathbb{S}^2$

Method

- ▶ Measure the traveltimes of surface waves between many pairs of epicenter and detector
- ▶ Reconstruct the local speed of propagation

Assumption

A wave propagates along the arc of a great circle.



P. Funk.

Über Flächen mit lauter geschlossenen geodätischen Linien.

Math. Ann., 74(2): 278 – 300, 1913.



J. H. Woodhouse and A. M. Dziewonski.

Mapping the upper mantle: Three-dimensional modeling of earth structure by inversion of seismic waveforms.

J. Geophys. Res. Solid Earth, 89(B7):5953–5986, 1984.



A. Amirbekyan, V. Michel, and F. J. Simons.

Parametrizing surface wave tomographic models with harmonic spherical splines.

Geophys. J. Int., 174(2):617–628, 2008.



R. Hielscher, D. Potts and M. Quellmalz.

An SVD in spherical surface wave tomography

In B. Hofmann, A. Leitao and J. Zubelli, Eds., *New Trends in Parameter Identification for Mathematical Models*. Birkhäuser, Basel, 2018.

<https://arxiv.org/abs/1706.05284>



P. Funk.

Über Flächen mit lauter geschlossenen geodätischen Linien.

Math. Ann., 74(2): 278 – 300, 1913.



J. H. Woodhouse and A. M. Dziewonski.

Mapping the upper mantle: Three-dimensional modeling of earth structure by inversion of seismic waveforms.

J. Geophys. Res. Solid Earth, 89(B7):5953–5986, 1984.



A. Amirbekyan, V. Michel, and F. J. Simons.

Parametrizing surface wave tomographic models with harmonic spherical splines.

Geophys. J. Int., 174(2):617–628, 2008.



R. Hielscher, D. Potts and M. Quellmalz.

An SVD in spherical surface wave tomography

In B. Hofmann, A. Leitao and J. Zubelli, Eds., *New Trends in Parameter Identification for Mathematical Models*. Birkhäuser, Basel, 2018.

<https://arxiv.org/abs/1706.05284>



P. Funk.

Über Flächen mit lauter geschlossenen geodätischen Linien.

Math. Ann., 74(2): 278 – 300, 1913.



J. H. Woodhouse and A. M. Dziewonski.

Mapping the upper mantle: Three-dimensional modeling of earth structure by inversion of seismic waveforms.

J. Geophys. Res. Solid Earth, 89(B7):5953–5986, 1984.



A. Amirbekyan, V. Michel, and F. J. Simons.

Parametrizing surface wave tomographic models with harmonic spherical splines.

Geophys. J. Int., 174(2):617–628, 2008.



R. Hielscher, D. Potts and M. Quellmalz.

An SVD in spherical surface wave tomography

In B. Hofmann, A. Leitao and J. Zubelli, Eds., *New Trends in Parameter Identification for Mathematical Models*. Birkhäuser, Basel, 2018.

<https://arxiv.org/abs/1706.05284>



P. Funk.

Über Flächen mit lauter geschlossenen geodätischen Linien.

Math. Ann., 74(2): 278 – 300, 1913.



J. H. Woodhouse and A. M. Dziewonski.

Mapping the upper mantle: Three-dimensional modeling of earth structure by inversion of seismic waveforms.

J. Geophys. Res. Solid Earth, 89(B7):5953–5986, 1984.



A. Amirbekyan, V. Michel, and F. J. Simons.

Parametrizing surface wave tomographic models with harmonic spherical splines.

Geophys. J. Int., 174(2):617–628, 2008.



R. Hielscher, D. Potts and M. Quellmalz.

An SVD in spherical surface wave tomography

In B. Hofmann, A. Leitao and J. Zubelli, Eds., *New Trends in Parameter Identification for Mathematical Models*. Birkhäuser, Basel, 2018.

<https://arxiv.org/abs/1706.05284>

Content

1. Introduction

Motivation

2. Arc transform

Definition

Singular value decomposition

3. Special cases

Arcs starting in a fixed point

Recovery of local functions

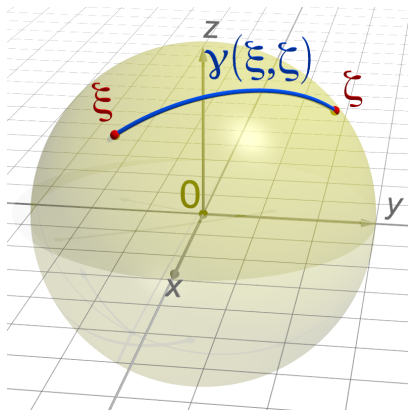
Arcs with fixed length

The arc transform

- ▶ **Function $f: \mathbb{S}^2 \rightarrow \mathbb{R}$**
 - ▶ Surface waves: $f = \frac{1}{c}$
(c ... speed of sound)
- ▶ $\xi, \zeta \in \mathbb{S}^2$ not antipodal
- ▶ $\gamma(\xi, \zeta)$ great circle arc

Definition

$$t(\xi, \zeta) = \int_{\gamma(\xi, \zeta)} f \, d\gamma$$

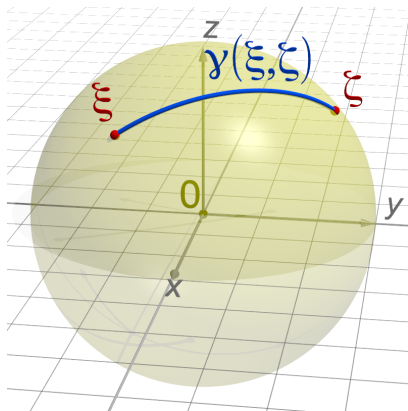


The arc transform

- ▶ Function $f: \mathbb{S}^2 \rightarrow \mathbb{R}$
 - ▶ Surface waves: $f = \frac{1}{c}$
(c ... speed of sound)
- ▶ $\xi, \zeta \in \mathbb{S}^2$ not antipodal
- ▶ $\gamma(\xi, \zeta)$ great circle arc

Definition

$$t(\xi, \zeta) = \int_{\gamma(\xi, \zeta)} f \, d\gamma$$

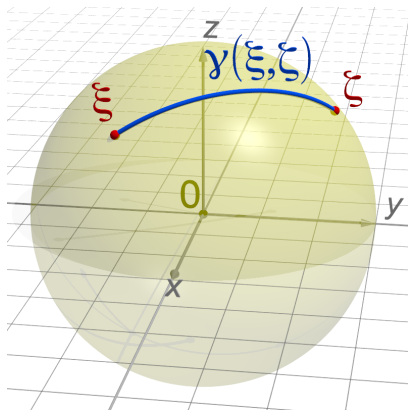


The arc transform

- ▶ Function $f: \mathbb{S}^2 \rightarrow \mathbb{R}$
 - ▶ Surface waves: $f = \frac{1}{c}$
(c ... speed of sound)
- ▶ $\xi, \zeta \in \mathbb{S}^2$ not antipodal
- ▶ $\gamma(\xi, \zeta)$ great circle arc

Definition

$$t(\xi, \zeta) = \int_{\gamma(\xi, \zeta)} f \, d\gamma$$

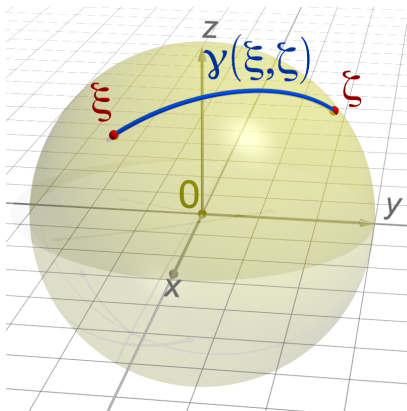


The arc transform

- ▶ Function $f: \mathbb{S}^2 \rightarrow \mathbb{R}$
 - ▶ Surface waves: $f = \frac{1}{c}$
(c ... speed of sound)
- ▶ $\xi, \zeta \in \mathbb{S}^2$ not antipodal
- ▶ $\gamma(\xi, \zeta)$ great circle arc

Definition

$$t(\xi, \zeta) = \int_{\gamma(\xi, \zeta)} f \, d\gamma$$

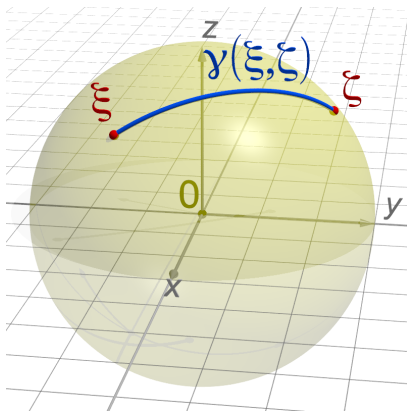


The arc transform

- ▶ Function $f: \mathbb{S}^2 \rightarrow \mathbb{R}$
 - ▶ Surface waves: $f = \frac{1}{c}$
(c ... speed of sound)
- ▶ $\xi, \zeta \in \mathbb{S}^2$ not antipodal
- ▶ $\gamma(\xi, \zeta)$ great circle arc

Definition

$$t(\xi, \zeta) = \int_{\gamma(\xi, \zeta)} f \, d\gamma$$



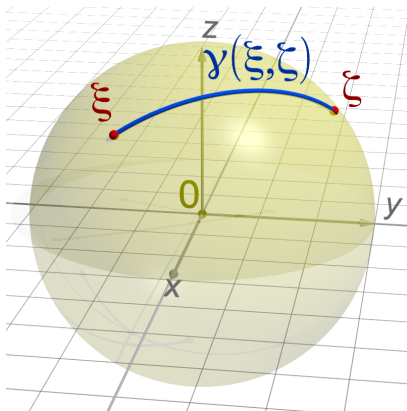
The arc transform

- ▶ Function $f: \mathbb{S}^2 \rightarrow \mathbb{R}$
 - ▶ Surface waves: $f = \frac{1}{c}$
($c \dots$ speed of sound)
- ▶ $\xi, \zeta \in \mathbb{S}^2$ not antipodal
- ▶ $\gamma(\xi, \zeta)$ great circle arc

Definition

$$t(\xi, \zeta) = \int_{\gamma(\xi, \zeta)} f \, d\gamma$$

t is not continuous on $\mathbb{S}^2 \times \mathbb{S}^2$



The arc transform

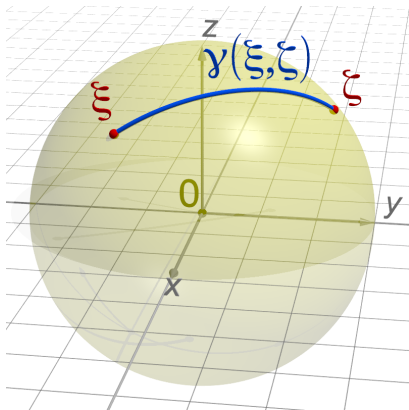
- ▶ Function $f: \mathbb{S}^2 \rightarrow \mathbb{R}$
 - ▶ Surface waves: $f = \frac{1}{c}$
(c ... speed of sound)
- ▶ $\xi, \zeta \in \mathbb{S}^2$ not antipodal
- ▶ $\gamma(\xi, \zeta)$ great circle arc

Definition

$$t(\xi, \zeta) = \int_{\gamma(\xi, \zeta)} f \, d\gamma$$

t is not continuous on $\mathbb{S}^2 \times \mathbb{S}^2$

We choose a different parameterization



The arc transform: alternative parameterization

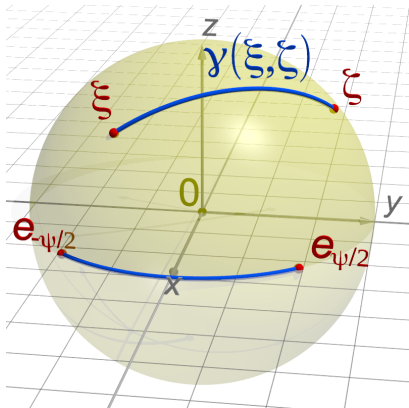
- ▶ $\psi = \arccos(\xi^\top \zeta)$... length of γ
- ▶ $Q \in \text{SO}(3)$ such that
 - ▶ $Q\xi = e_{-\psi/2}$ and
 - ▶ $Q\zeta = e_{\psi/2}$,

where $e_\psi = (\sin \psi, \cos \psi, 0)$

Definition

$\mathcal{A}: C(\mathbb{S}^2) \rightarrow C(\text{SO}(3) \times [0, 2\pi])$,

$$\mathcal{A}f(Q, \psi) = \int_{-\psi/2}^{\psi/2} f(Q^{-1}e_\varphi) d\varphi$$



The arc transform: alternative parameterization

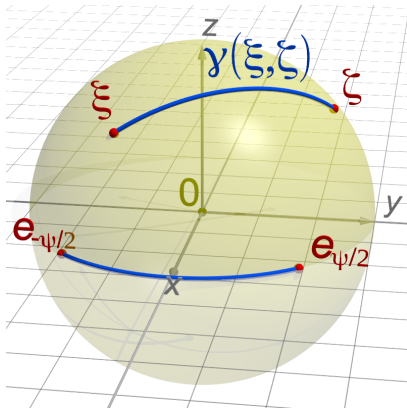
- ▶ $\psi = \arccos(\xi^\top \zeta)$... length of γ
- ▶ $Q \in \text{SO}(3)$ such that
 - ▶ $Q\xi = e_{-\psi/2}$ and
 - ▶ $Q\zeta = e_{\psi/2}$,

where $e_\psi = (\sin \psi, \cos \psi, 0)$

Definition

$$\mathcal{A}: C(\mathbb{S}^2) \rightarrow C(\text{SO}(3) \times [0, 2\pi]),$$

$$\mathcal{A}f(Q, \psi) = \int_{-\psi/2}^{\psi/2} f(Q^{-1}e_\varphi) d\varphi$$



The arc transform: alternative parameterization

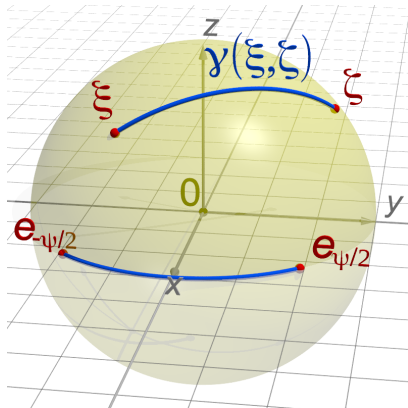
- ▶ $\psi = \arccos(\xi^\top \zeta) \dots$ length of γ
- ▶ $Q \in \text{SO}(3)$ such that
 - ▶ $Q\xi = e_{-\psi/2}$ and
 - ▶ $Q\zeta = e_{\psi/2}$,

where $e_\psi = (\sin \psi, \cos \psi, 0)$

Definition

$$\mathcal{A}: C(\mathbb{S}^2) \rightarrow C(\text{SO}(3) \times [0, 2\pi]),$$

$$\mathcal{A}f(Q, \psi) = \int_{-\psi/2}^{\psi/2} f(Q^{-1}e_\varphi) d\varphi$$



On the sphere $\mathbb{S}^2$

► Polar coordinates

$$\xi(\varphi, t) = \sqrt{1 - t^2} \, e_\varphi + t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

► Orthonormal basis on $L^2(\mathbb{S}^2)$: **spherical harmonics** of degree $n \in \mathbb{N}$

$$Y_n^k(\varphi, t) = \sqrt{\frac{2n+1}{4\pi} \frac{(n-k)!}{(n+k)!}} P_n^k(t) e^{ik\varphi}, \quad k = -n, \dots, n$$

► P_n^k ... associated Legendre function

On the sphere $\mathbb{S}^2$

- Polar coordinates

$$\xi(\varphi, t) = \sqrt{1 - t^2} \, e_\varphi + t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

- Orthonormal basis on $L^2(\mathbb{S}^2)$: **spherical harmonics** of degree $n \in \mathbb{N}$

$$Y_n^k(\varphi, t) = \sqrt{\frac{2n+1}{4\pi} \frac{(n-k)!}{(n+k)!}} P_n^k(t) e^{ik\varphi}, \quad k = -n, \dots, n$$

- P_n^k ... associated Legendre function

On the rotation group $SO(3)$

► Rotation group

$$SO(3) = \{Q \in \mathbb{R}^{3 \times 3} : Q^{-1} = Q^T, \det(Q) = 1\}$$

- Orthogonal basis on $L^2(SO(3))$: **rotational harmonics** (Wigner D-functions)

$$D_n^{j,k}(Q) = \int_{\mathbb{S}^2} Y_n^k(Q^{-1}\xi) \overline{Y_n^j(\xi)} \, d\xi$$

On the rotation group $SO(3)$

► Rotation group

$$SO(3) = \{Q \in \mathbb{R}^{3 \times 3} : Q^{-1} = Q^{\top}, \det(Q) = 1\}$$

► Orthogonal basis on $L^2(SO(3))$: **rotational harmonics** (Wigner D-functions)

$$D_n^{j,k}(Q) = \int_{\mathbb{S}^2} Y_n^k(Q^{-1}\xi) \overline{Y_n^j(\xi)} d\xi$$

Theorem

[Dahlen & Tromp 1998]

Let $n \in \mathbb{N}$ and $k \in \{-n, \dots, n\}$. Then

$$AY_n^k(Q, \psi) = \sum_{j=-n}^n \tilde{P}_n^j(0) D_n^{j,k}(Q) s_j(\psi),$$

where

$$s_j(\psi) = \begin{cases} \psi, & j = 0 \\ \frac{2 \sin(j\psi/2)}{j}, & j \neq 0 \end{cases}$$

and

$$\tilde{P}_n^j(0) = \begin{cases} (-1)^{\frac{n+j}{2}} \sqrt{\frac{2n+1}{4\pi} \frac{(n-j-1)!!(n+j-1)!!}{(n-j)!!(n+j)!!}}, & n+j \text{ even} \\ 0, & n+j \text{ odd.} \end{cases}$$

Singular value decomposition

[Hielscher, Potts, Q. 2017]

The operator $\mathcal{A}: L^2(\mathbb{S}^2) \rightarrow L^2(\mathrm{SO}(3) \times [0, 2\pi])$ is compact with the singular value decomposition

$$\{(Y_n^k, E_n^k, \sigma_n) : n \in \mathbb{N}, k \in \{-n, \dots, n\}\},$$

with singular values

$$\sigma_n = \sqrt{\frac{32\pi^3}{2n+1}} \sqrt{\frac{\pi^2}{3} |\tilde{P}_n^0(0)|^2 + \sum_{j=1}^n \frac{1}{j^2} |\tilde{P}_n^j(0)|^2} \in \mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$$

and the orthonormal functions in $L^2(\mathrm{SO}(3) \times [0, 2\pi])$

$$E_k^n = \sigma_n^{-1} \sum_{j=-n}^n \tilde{P}_n^j(0) D_n^{j,k}(Q) s_j(\psi).$$

Content

1. Introduction

Motivation

2. Arc transform

Definition

Singular value decomposition

3. Special cases

Arcs starting in a fixed point

Recovery of local functions

Arcs with fixed length

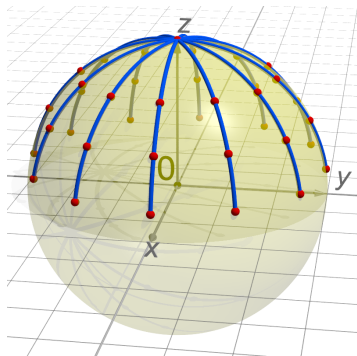
Arcs from the north pole

- Fix one endpoint of the arcs as the north pole e^3 :

$$\mathcal{B}f(\xi(\varphi, \vartheta)) = \int_{\gamma(e^3, \xi(\varphi, \vartheta))} f \, d\gamma$$

- If f is differentiable, it can be recovered from $\mathcal{B}f$ by

$$f(\xi(\varphi, \vartheta)) = \frac{d}{d\vartheta} \mathcal{B}f(\xi(\varphi, \vartheta)).$$



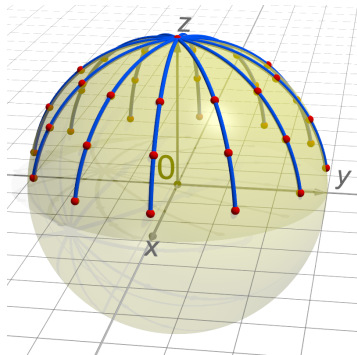
Arcs from the north pole

- Fix one endpoint of the arcs as the north pole e^3 :

$$\mathcal{B}f(\xi(\varphi, \vartheta)) = \int_{\gamma(e^3, \xi(\varphi, \vartheta))} f \, d\gamma$$

- If f is differentiable, it can be recovered from $\mathcal{B}f$ by

$$f(\xi(\varphi, \vartheta)) = \frac{d}{d\vartheta} \mathcal{B}f(\xi(\varphi, \vartheta)).$$



Arcs between two sets

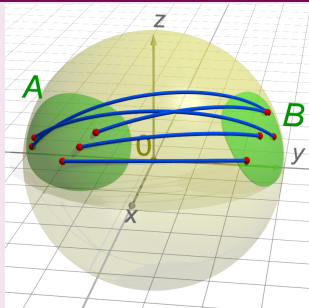
More general Theorem

[Amirbekyan 2007]

Let S be an open subset of $\mathbb{S}^2$ and $A, B \subset S$ nonempty sets with $\overline{A \cup B} = \overline{S}$. If $f \in C(\mathbb{S}^2)$ and

$$\int_{\gamma(\xi, \zeta)} f \, d\gamma = 0 \quad \forall \xi \in A, \zeta \in B,$$

then $f \equiv 0$ on S .



Theorem

[Hielscher, Potts, Q. 2017]

Let $f \in C(\mathbb{S}^2)$ and Ω be a convex subset of $\mathbb{S}^2$ whose closure $\overline{\Omega}$ is strictly contained in a hemisphere, i.e., there exists a $\zeta \in \mathbb{S}^2$ such that $\langle \xi, \zeta \rangle > 0$ for all $\xi \in \overline{\Omega}$. If

$$\int_{\gamma(\xi, \eta)} f \, d\gamma = 0 \quad \text{for all } \xi, \eta \in \partial\Omega, \quad (1)$$

then $f = 0$ on Ω .

Proof

- Extend f to zero outside Ω
- (1) implies that the Funk–Radon transform of f must vanish
- f must be odd
- Because $\text{supp } f \subset \overline{\Omega}$ is contained in a hemisphere, f must vanish

Theorem

[Hielscher, Potts, Q. 2017]

Let $f \in C(\mathbb{S}^2)$ and Ω be a convex subset of $\mathbb{S}^2$ whose closure $\overline{\Omega}$ is strictly contained in a hemisphere, i.e., there exists a $\zeta \in \mathbb{S}^2$ such that $\langle \xi, \zeta \rangle > 0$ for all $\xi \in \overline{\Omega}$. If

$$\int_{\gamma(\xi, \eta)} f \, d\gamma = 0 \quad \text{for all } \xi, \eta \in \partial\Omega, \quad (1)$$

then $f = 0$ on Ω .

Proof

- ▶ Extend f to zero outside Ω
- ▶ (1) implies that the Funk–Radon transform of f must vanish
- ▶ f must be odd
- ▶ Because $\text{supp } f \subset \overline{\Omega}$ is contained in a hemisphere, f must vanish

Theorem

[Hielscher, Potts, Q. 2017]

Let $f \in C(\mathbb{S}^2)$ and Ω be a convex subset of $\mathbb{S}^2$ whose closure $\overline{\Omega}$ is strictly contained in a hemisphere, i.e., there exists a $\zeta \in \mathbb{S}^2$ such that $\langle \xi, \zeta \rangle > 0$ for all $\xi \in \overline{\Omega}$. If

$$\int_{\gamma(\xi, \eta)} f \, d\gamma = 0 \quad \text{for all } \xi, \eta \in \partial\Omega, \quad (1)$$

then $f = 0$ on Ω .

Proof

- ▶ Extend f to zero outside Ω
- ▶ (1) implies that the Funk–Radon transform of f must vanish
- ▶ f must be odd
- ▶ Because $\text{supp } f \subset \overline{\Omega}$ is contained in a hemisphere, f must vanish

Theorem

[Hielscher, Potts, Q. 2017]

Let $f \in C(\mathbb{S}^2)$ and Ω be a convex subset of $\mathbb{S}^2$ whose closure $\overline{\Omega}$ is strictly contained in a hemisphere, i.e., there exists a $\zeta \in \mathbb{S}^2$ such that $\langle \xi, \zeta \rangle > 0$ for all $\xi \in \overline{\Omega}$. If

$$\int_{\gamma(\xi, \eta)} f \, d\gamma = 0 \quad \text{for all } \xi, \eta \in \partial\Omega, \quad (1)$$

then $f = 0$ on Ω .

Proof

- ▶ Extend f to zero outside Ω
- ▶ (1) implies that the Funk–Radon transform of f must vanish
- ▶ f must be odd
- ▶ Because $\text{supp } f \subset \overline{\Omega}$ is contained in a hemisphere, f must vanish

Theorem

[Hielscher, Potts, Q. 2017]

Let $f \in C(\mathbb{S}^2)$ and Ω be a convex subset of $\mathbb{S}^2$ whose closure $\overline{\Omega}$ is strictly contained in a hemisphere, i.e., there exists a $\zeta \in \mathbb{S}^2$ such that $\langle \xi, \zeta \rangle > 0$ for all $\xi \in \overline{\Omega}$. If

$$\int_{\gamma(\xi, \eta)} f \, d\gamma = 0 \quad \text{for all } \xi, \eta \in \partial\Omega, \quad (1)$$

then $f = 0$ on Ω .

Proof

- ▶ Extend f to zero outside Ω
- ▶ (1) implies that the Funk–Radon transform of f must vanish
- ▶ f must be odd
- ▶ Because $\text{supp } f \subset \overline{\Omega}$ is contained in a hemisphere, f must vanish

Theorem

[Hielscher, Potts, Q. 2017]

Let $f \in C(\mathbb{S}^2)$ and Ω be a convex subset of $\mathbb{S}^2$ whose closure $\bar{\Omega}$ is strictly contained in a hemisphere, i.e., there exists a $\zeta \in \mathbb{S}^2$ such that $\langle \xi, \zeta \rangle > 0$ for all $\xi \in \bar{\Omega}$. If

$$\int_{\gamma(\xi, \eta)} f \, d\gamma = 0 \quad \text{for all } \xi, \eta \in \partial\Omega, \quad (1)$$

then $f = 0$ on Ω .

Proof

- ▶ Extend f to zero outside Ω
- ▶ (1) implies that the Funk–Radon transform of f must vanish
- ▶ f must be odd
- ▶ Because $\text{supp } f \subset \bar{\Omega}$ is contained in a hemisphere, f must vanish

Arcs with fixed length

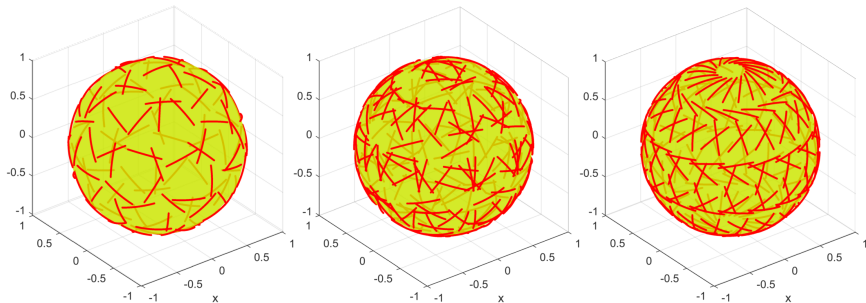
We fix the arclength $\psi \in [0, 2\pi]$ and define

$$\mathcal{A}_\psi = \mathcal{A}(\cdot, \psi) : L^2(\mathbb{S}^2) \rightarrow L^2(\mathrm{SO}(3)).$$

Arcs with fixed length

We fix the arclength $\psi \in [0, 2\pi]$ and define

$$\mathcal{A}_\psi = \mathcal{A}(\cdot, \psi) : L^2(\mathbb{S}^2) \rightarrow L^2(\text{SO}(3)).$$



Singular Value Decomposition

[Hielscher, Potts, Q. 2017]

Let $\psi \in (0, 2\pi)$ be fixed. The operator $\mathcal{A}_\psi: L^2(\mathbb{S}^2) \rightarrow L^2(\text{SO}(3))$ has the SVD

$$\left\{ \left(Y_n^k, Z_{n,\psi}^k, \mu_n(\psi) \right) : n \in \mathbb{N}, k \in \{-n, \dots, n\} \right\},$$

with singular values

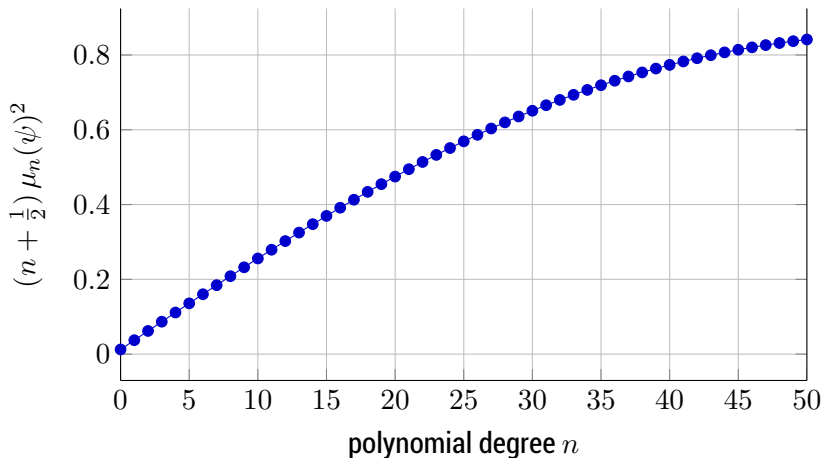
$$\mu_n(\psi) = \sqrt{\sum_{j=-n}^n \frac{8\pi^2}{2n+1} \left| \tilde{P}_n^j(0) \right|^2 s_j(\psi)^2}$$

and singular functions

$$Z_{n,\psi}^k = \frac{1}{\mu_n(\psi)} \sum_{j=-n}^n \tilde{P}_n^j(0) s_j(\psi) D_n^{j,k} \in L^2(\text{SO}(3)).$$

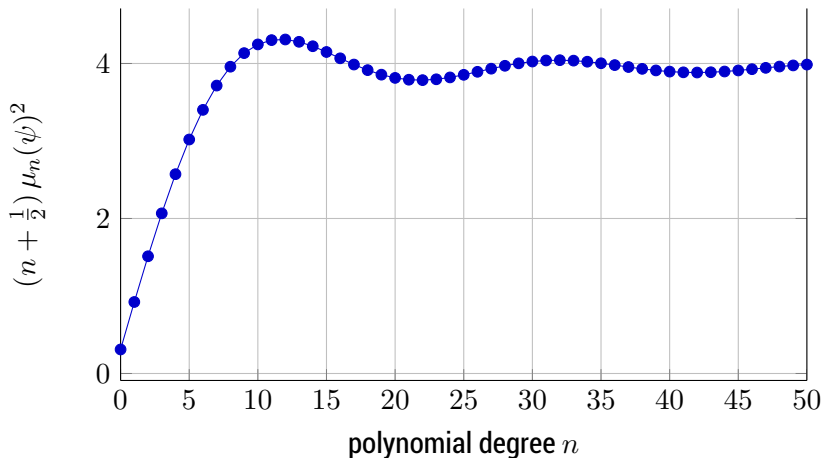
Singular values $\mu_n(\psi)$: dependency on n

$$\psi = 0.02\pi$$



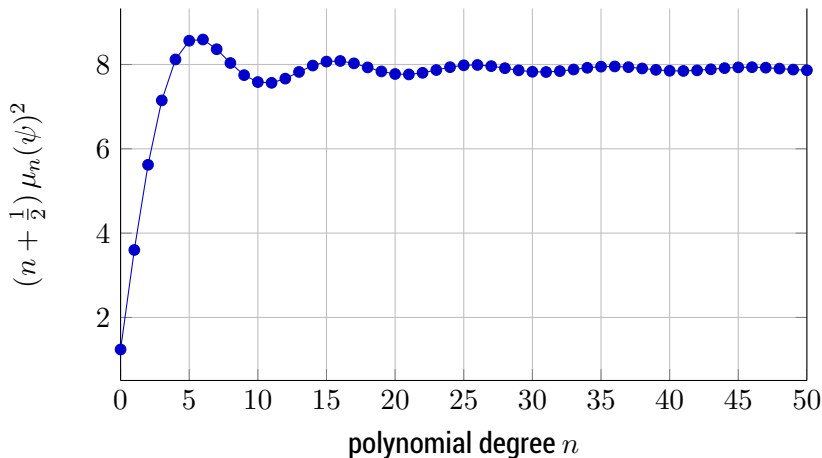
Singular values $\mu_n(\psi)$: dependency on n

$$\psi = 0.10\pi$$



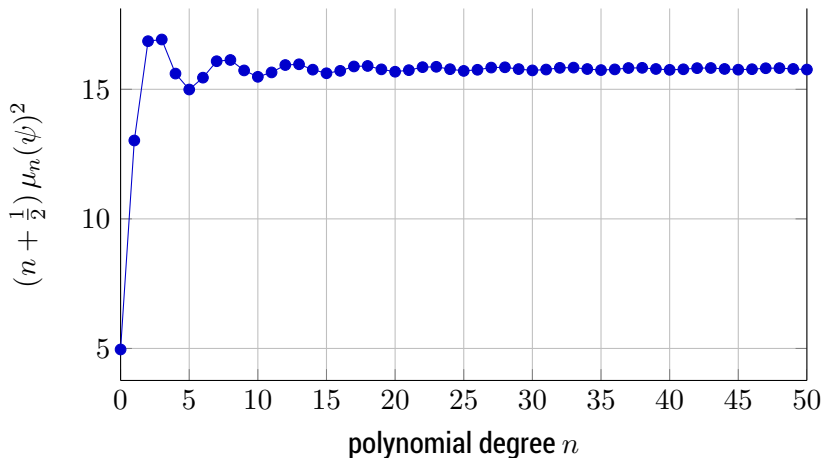
Singular values $\mu_n(\psi)$: dependency on n

$$\psi = 0.20\pi$$



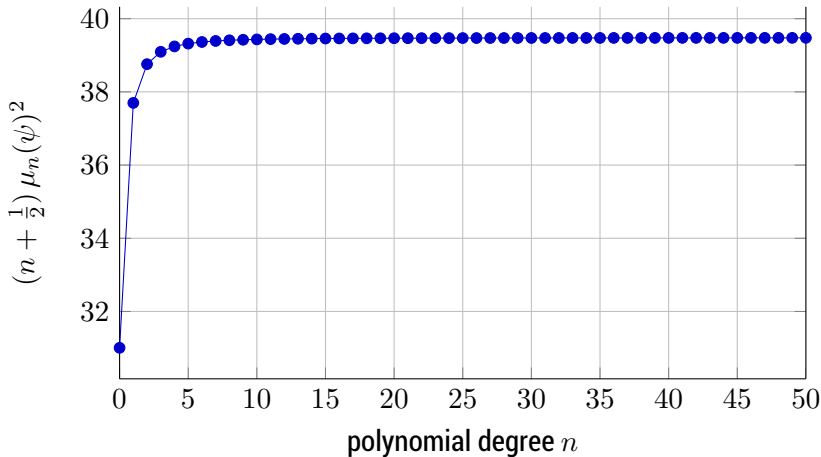
Singular values $\mu_n(\psi)$: dependency on n

$$\psi = 0.40\pi$$



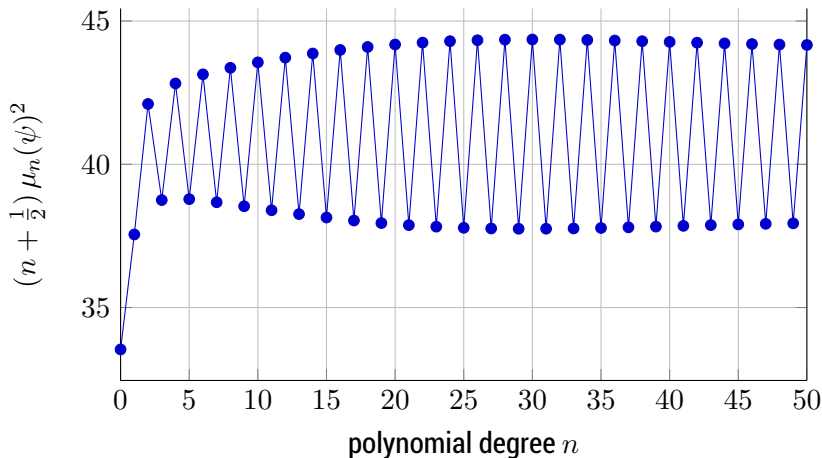
Singular values $\mu_n(\psi)$: dependency on n

$\psi = 1.00 \pi$ (half circle)



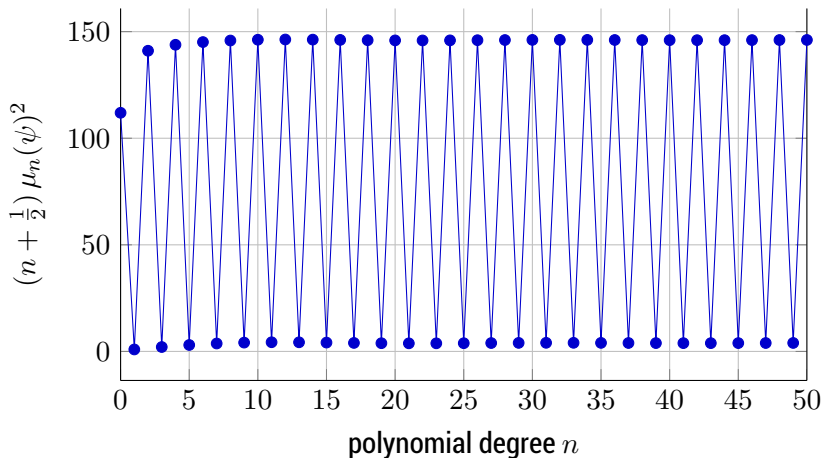
Singular values $\mu_n(\psi)$: dependency on n

$$\psi = 1.04\pi$$



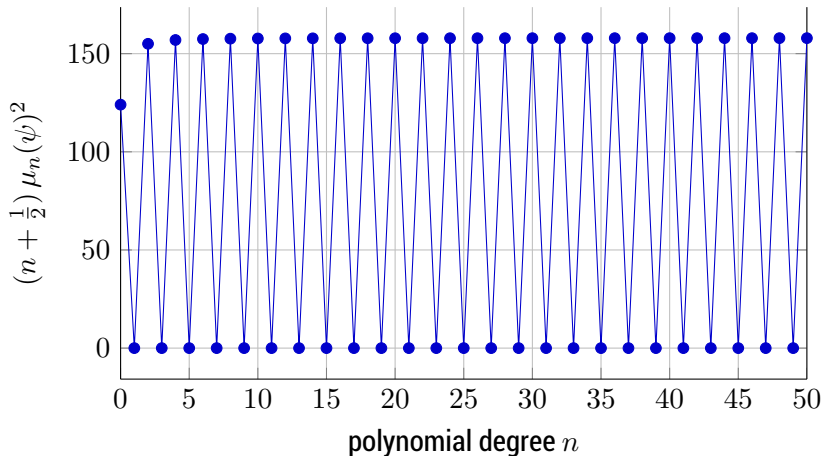
Singular values $\mu_n(\psi)$: dependency on n

$$\psi = 1.90\pi$$

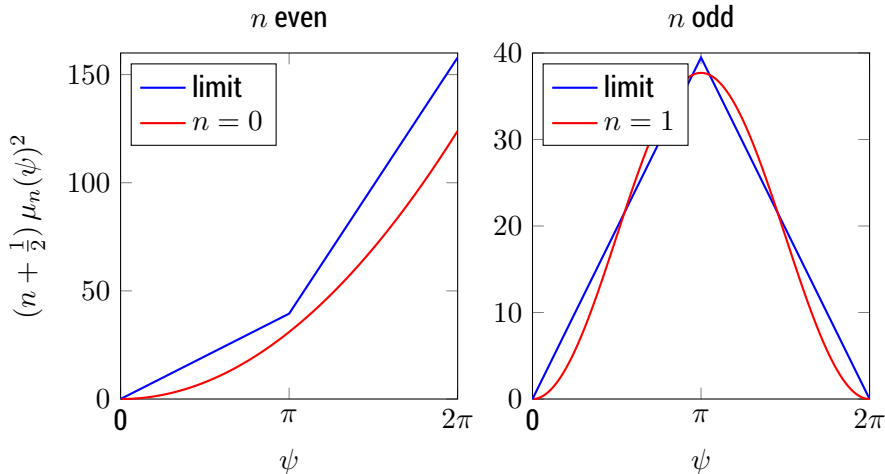


Singular values $\mu_n(\psi)$: dependency on n

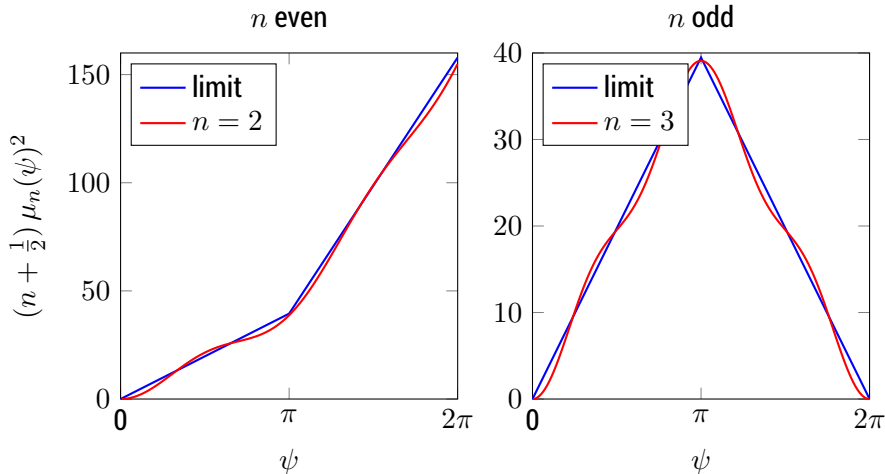
$\psi = 2.00 \pi$ (Funk–Radon transform)



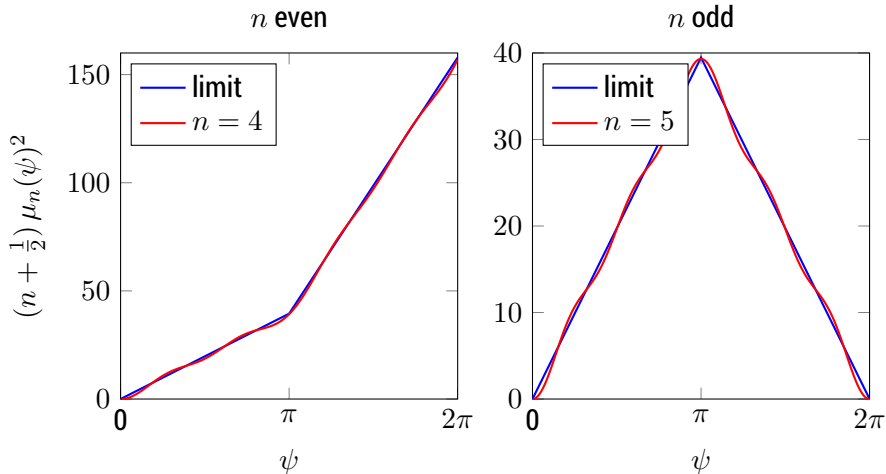
Singular values $\mu_n(\psi)$: dependency on arc-length ψ



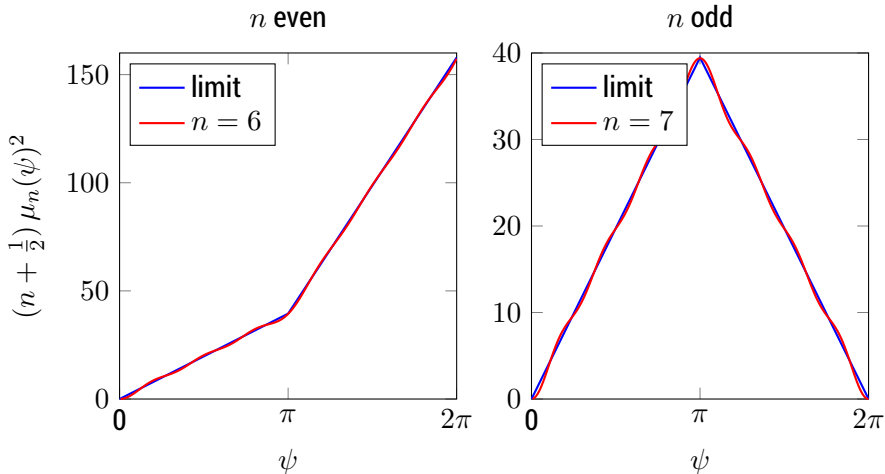
Singular values $\mu_n(\psi)$: dependency on arc-length ψ



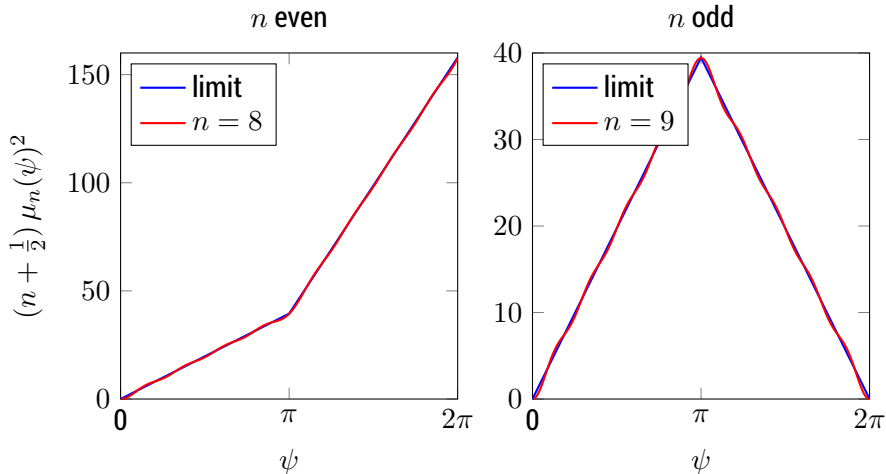
Singular values $\mu_n(\psi)$: dependency on arc-length ψ



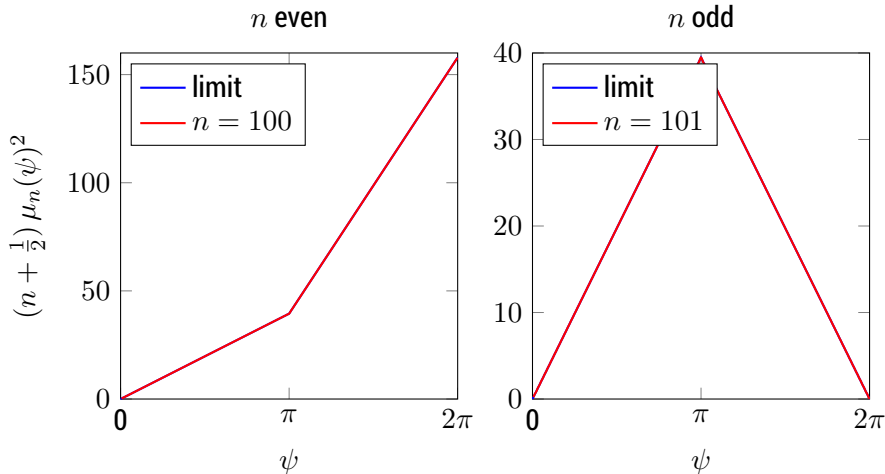
Singular values $\mu_n(\psi)$: dependency on arc-length ψ



Singular values $\mu_n(\psi)$: dependency on arc-length ψ



Singular values $\mu_n(\psi)$: dependency on arc-length ψ



Singular values: asymptotic behavior

Theorem

[Hielscher, Potts, Q. 2017]

The singular values $\mu_n(\psi)$ of $\mathcal{A}_\psi$ satisfy for odd $n = 2m - 1$

$$\lim_{m \rightarrow \infty} \frac{4m - 1}{4} \mu_{2m-1}(\psi)^2 = \begin{cases} 2\pi\psi, & \psi \in [0, \pi] \\ 4\pi^2 - 2\pi\psi, & \psi \in [\pi, 2\pi], \end{cases}$$

and for even $n = 2m$

$$\lim_{m \rightarrow \infty} \frac{4m + 1}{4} \mu_{2m}(\psi)^2 = \begin{cases} 2\pi\psi, & \psi \in [0, \pi] \\ 12\pi\psi - 2\pi^2, & \psi \in [\pi, 2\pi]. \end{cases}$$

$\psi = 2\pi$: Great circles

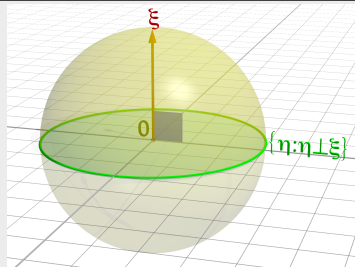
[Funk 1911]

► Funk–Radon transform

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) \, d\gamma(\eta)$$

► Injective only for even functions



$\psi = \pi$: Half-circle transform

- Injective for all functions [Groemer 1998], [Goodey & Weil 2006]
- [Rubin 2017] half circles in one hemisphere

$\psi = 2\pi$: Great circles

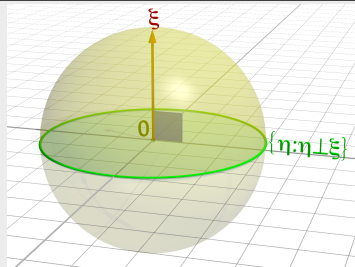
[Funk 1911]

► Funk–Radon transform

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) \, d\gamma(\eta)$$

► Injective only for even functions



$\psi = \pi$: Half-circle transform

- Injective for all functions [Groemer 1998], [Goodey & Weil 2006]
- [Rubin 2017] half circles in one hemisphere

$\psi = 2\pi$: Great circles

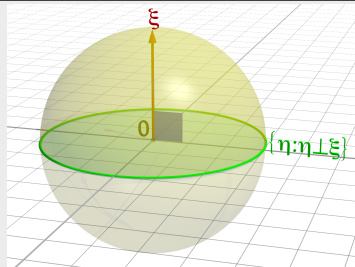
[Funk 1911]

► Funk–Radon transform

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) \, d\gamma(\eta)$$

► Injective only for even functions



$\psi = \pi$: Half-circle transform

- Injective for all functions [Groemer 1998], [Goodey & Weil 2006]
- [Rubin 2017] half circles in one hemisphere

$\psi = 2\pi$: Great circles

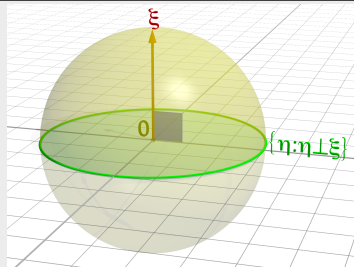
[Funk 1911]

► Funk–Radon transform

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) \, d\gamma(\eta)$$

► Injective only for even functions



$\psi = \pi$: Half-circle transform

- Injective for all functions [Groemer 1998], [Goodey & Weil 2006]
- [Rubin 2017] half circles in one hemisphere

$\psi = 2\pi$: Great circles

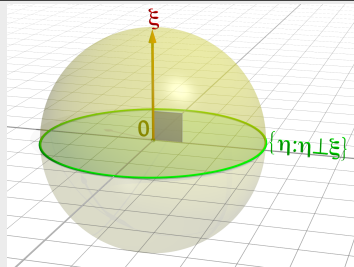
[Funk 1911]

► Funk–Radon transform

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) \, d\gamma(\eta)$$

► Injective only for even functions



$\psi = \pi$: Half-circle transform

- **Injective for all functions** [Groemer 1998], [Goodey & Weil 2006]
- [Rubin 2017] half circles in one hemisphere

$\psi = 2\pi$: Great circles

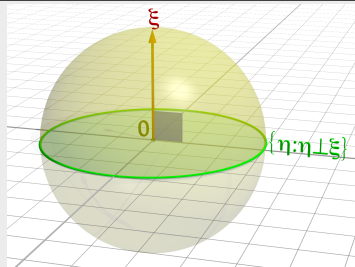
[Funk 1911]

► Funk–Radon transform

$$\mathcal{F}: C(\mathbb{S}^2) \rightarrow C(\mathbb{S}^2),$$

$$\mathcal{F}f(\xi) = \int_{\langle \xi, \eta \rangle = 0} f(\eta) \, d\gamma(\eta)$$

► Injective only for even functions



$\psi = \pi$: Half-circle transform

- **Injective for all functions** [Groemer 1998], [Goodey & Weil 2006]
- [Rubin 2017] half circles in one hemisphere

\endinput